INCH-POUND

MIL-DTL-24231/3E(SH)
w/AMENDMENT 1
13 May 2024
SUPERSEDING
MIL-DTL-24231/3E(SH)
26 April 2013

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, TYPE I, MOLDED, SEVEN- AND NINE-CONDUCTOR (STRAIGHT AND 90 DEGREES)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 7.

2. For termination of cable shield, see table for cable types for molded plugs in MIL-DTL-24231.

3. Sleeve M24231/3-010 may be modified prior to assembly by drilling or slotting to allow cable entry from the side. Sharp edges shall be rounded.

4. Ni-Cu plug sleeves shall be coated with plasma spray non-conductive material in accordance with S9320-AM-PRO-030/MLDG.

5. All dimensions are in inches.

6. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

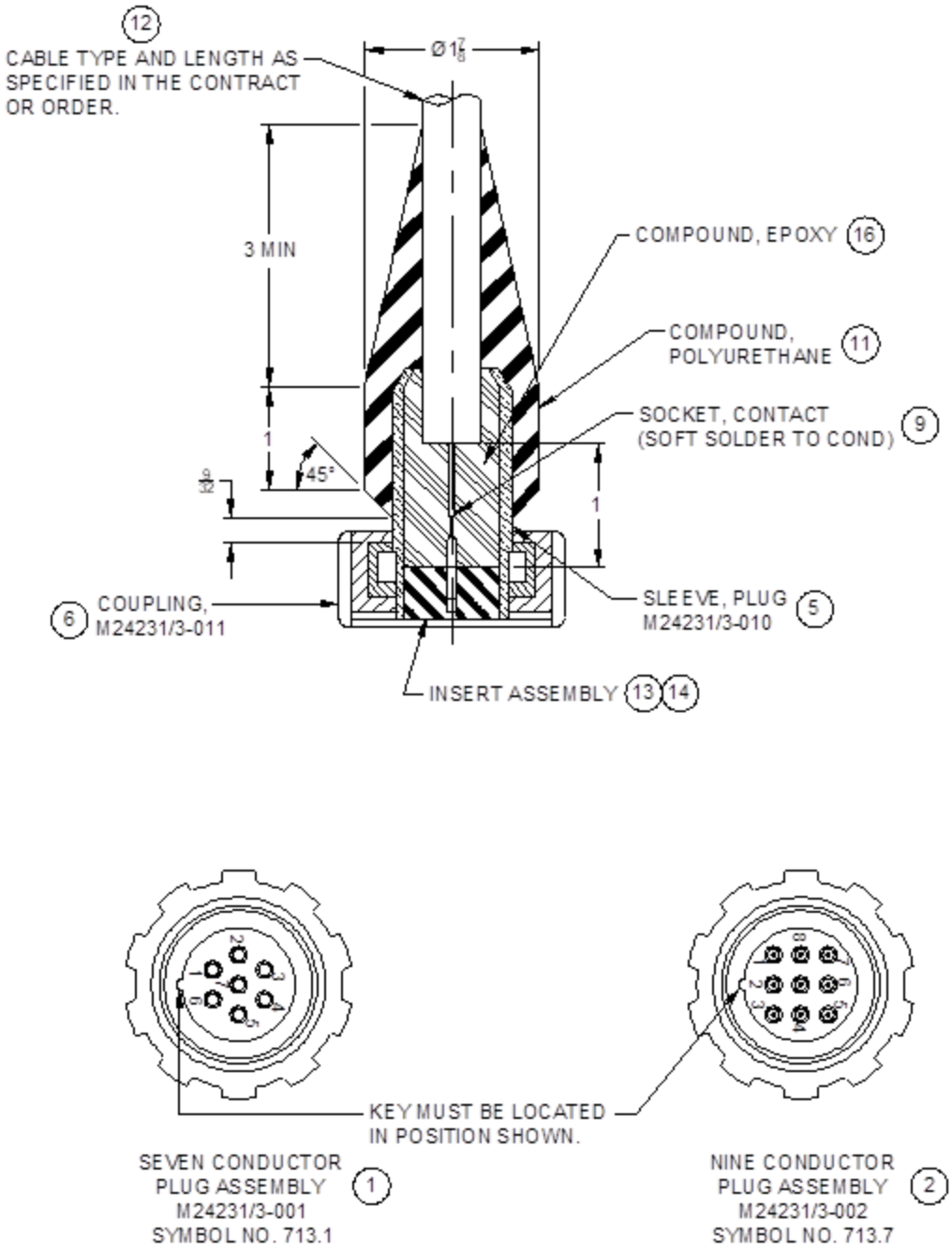


FIGURE 1. Connector, plug, molded, seven-, and nine-conductor (straight).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

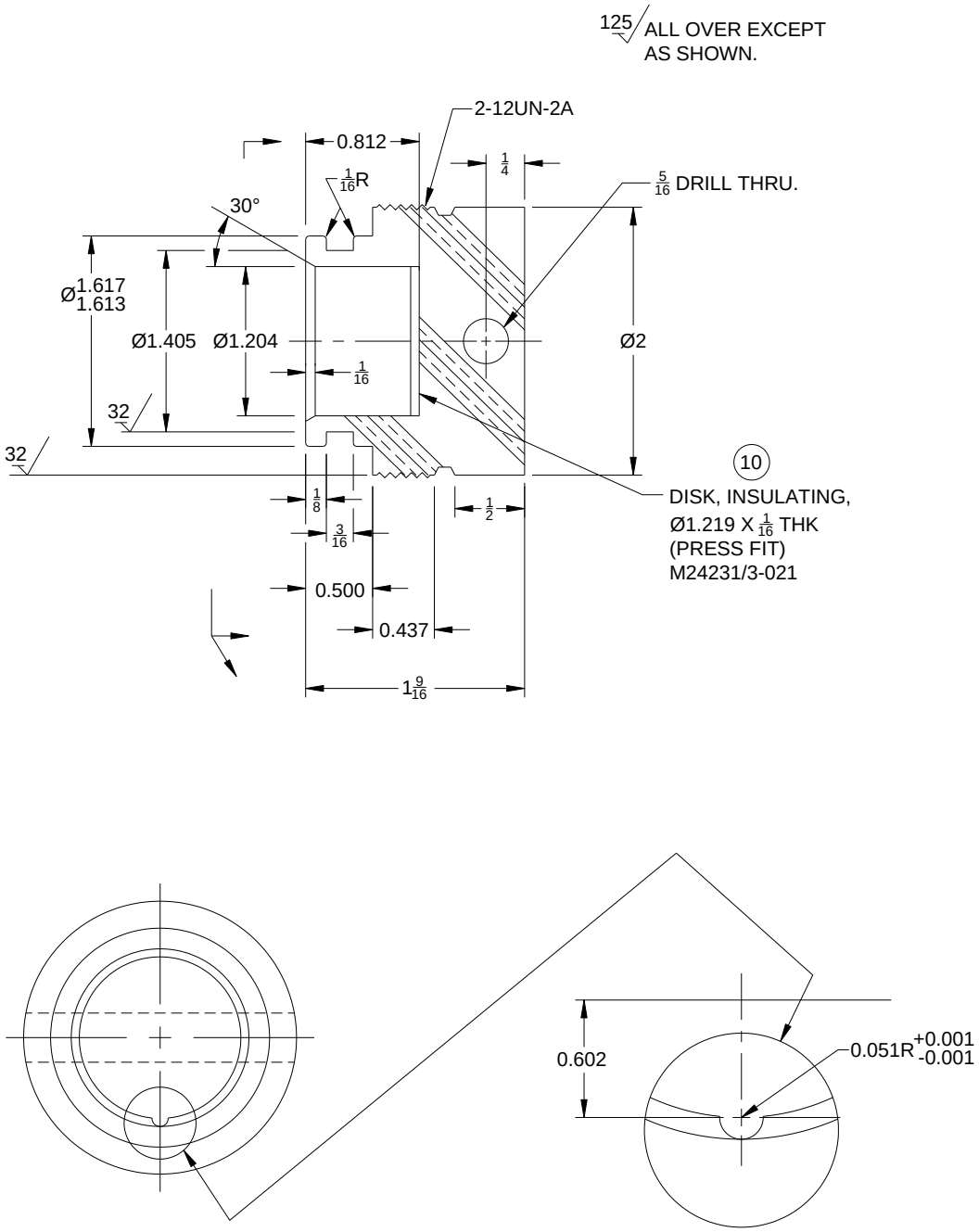


FIGURE 2. Sealing plug. (8)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

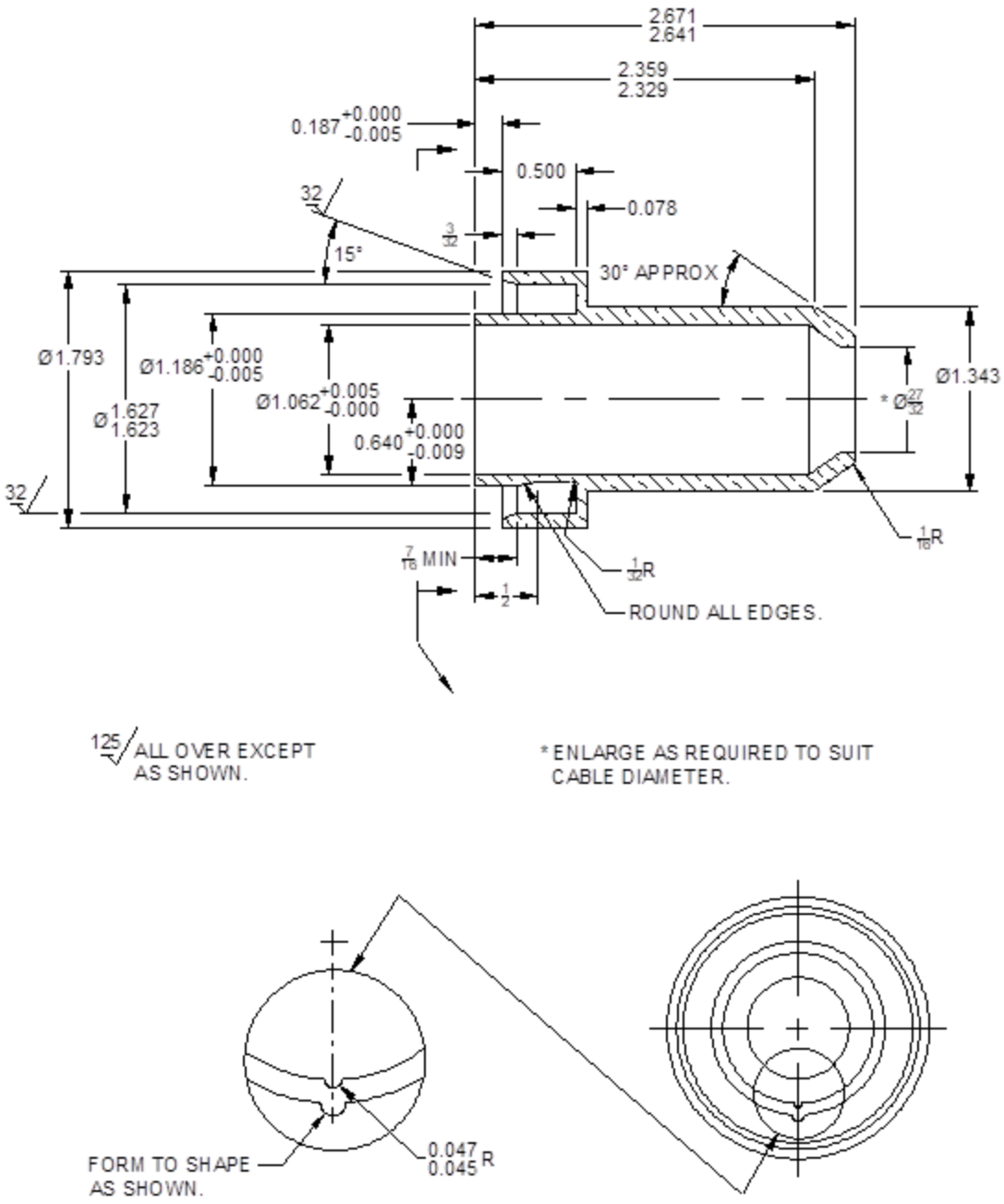


FIGURE 3. Plug sleeve – M24231/3-010. (5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

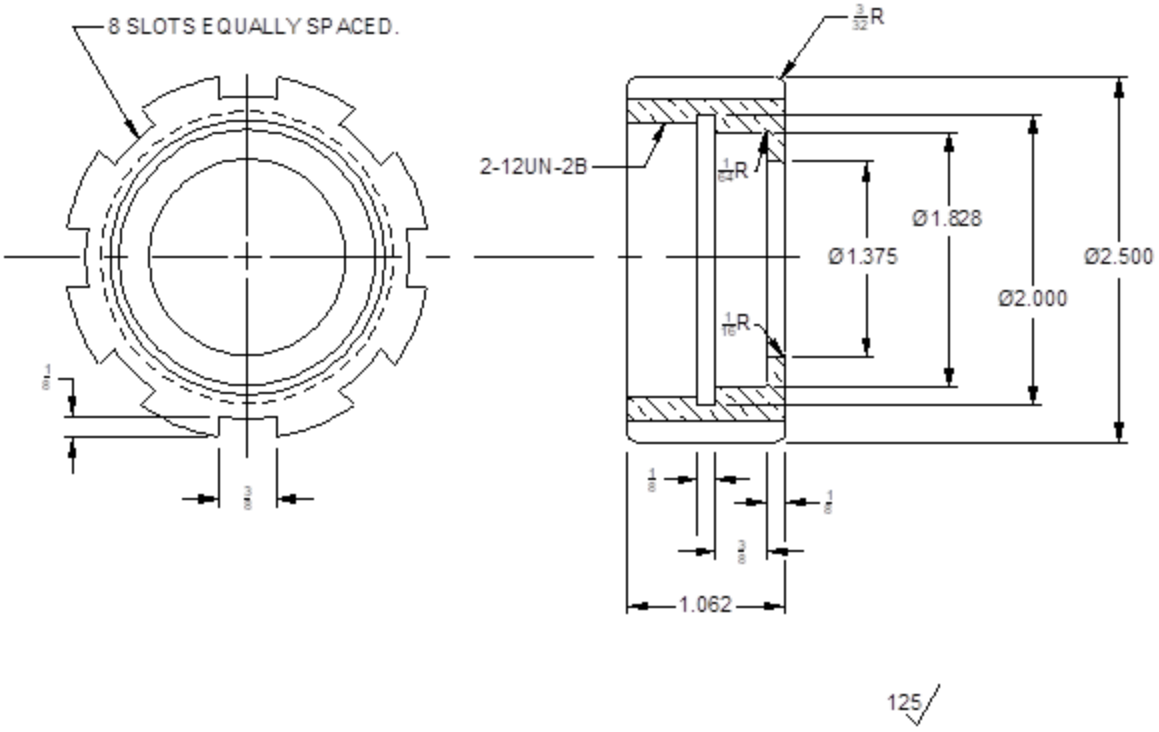


FIGURE 4. Coupling – M24231/3-011. (6)

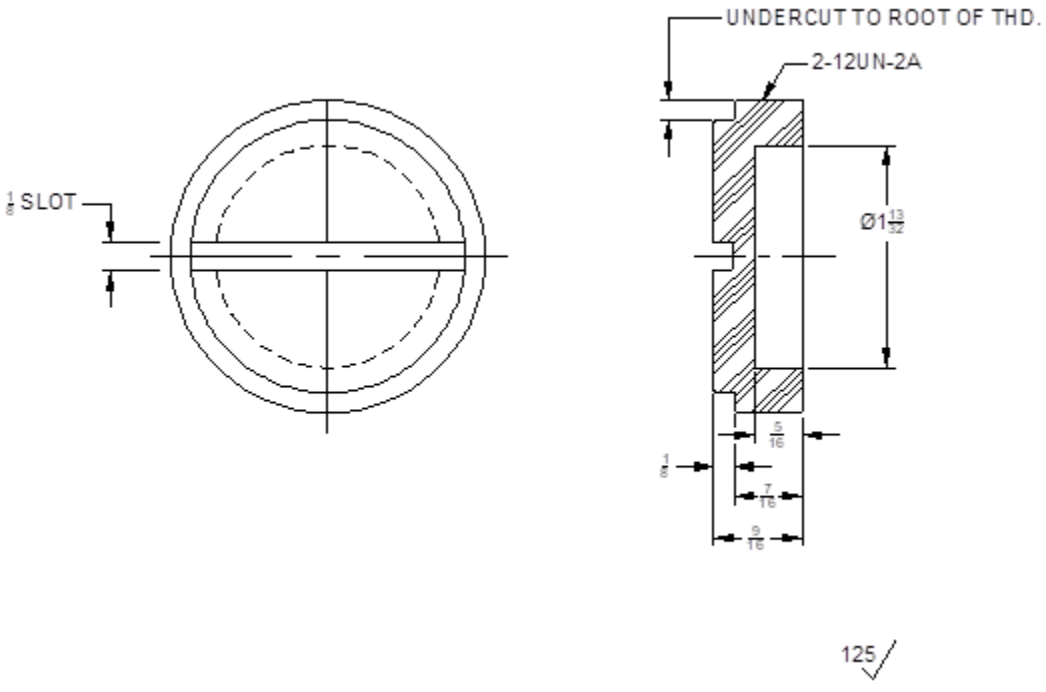


FIGURE 5. Cap, protective – M24231/3-012. (7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

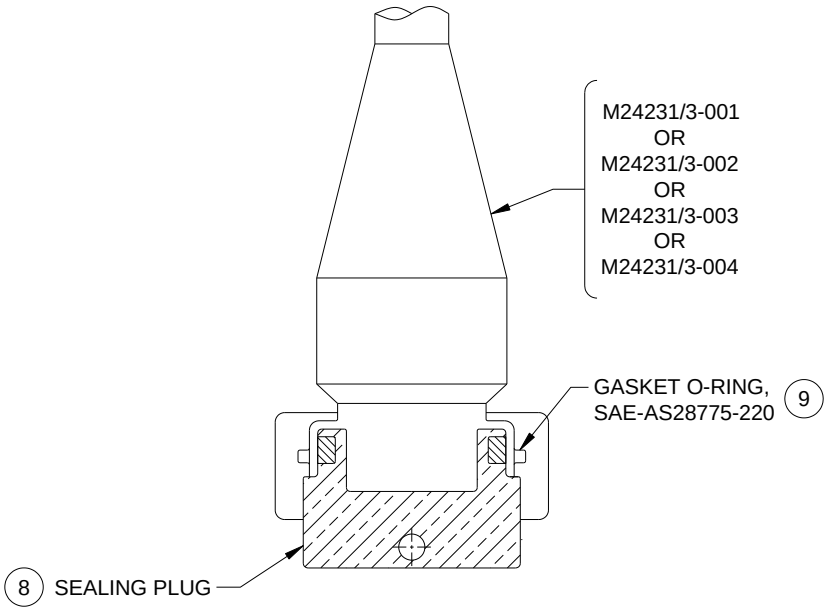
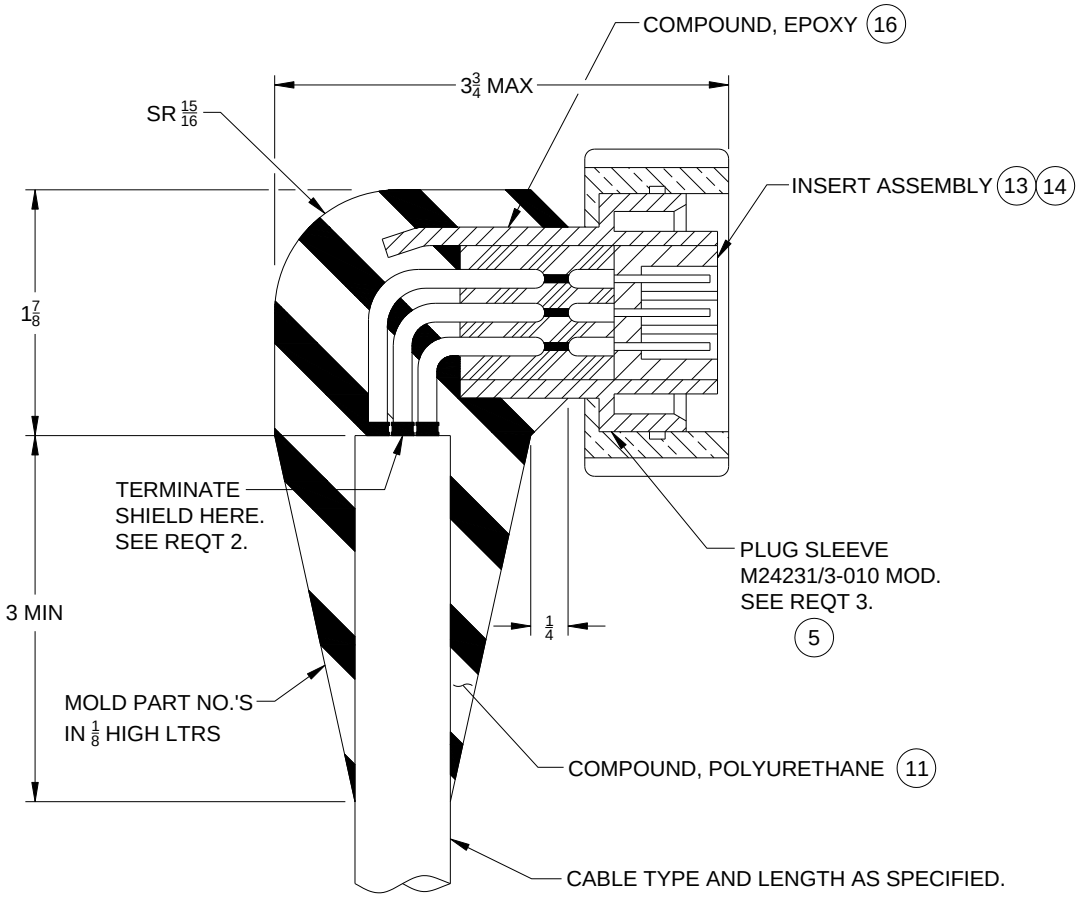


FIGURE 6. Sealing plug assembly – M24231/3-020. 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.



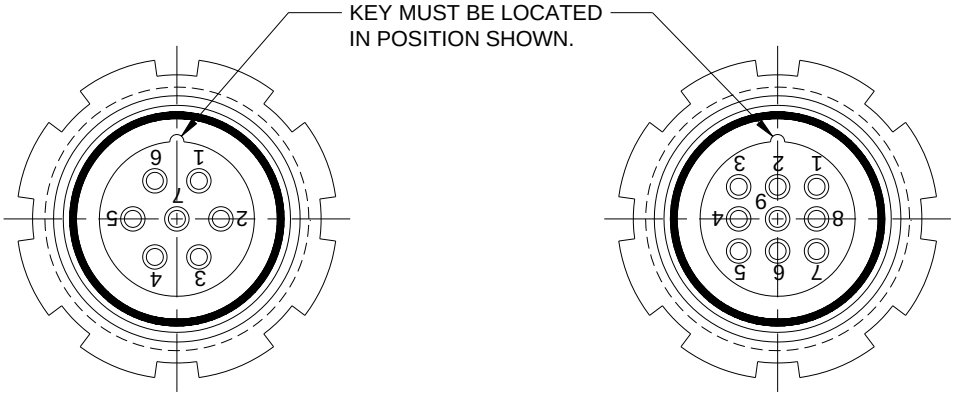


FIGURE 7. Connector, plug, molded, seven- and nine-conductor (90-degree).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list.

Item no. [1/]	Description [2/]	Quantity for M24231/3- [3/]				Material/specification	Remarks
		001	002	003	004		
1	Plug assembly, 7-conductor, symbol no. 713.1	1				[4/]	See figure 1.
2	Plug assembly, 9-conductor, symbol no. 713.7		1			[4/]	See figure 1.
3	Plug assembly, 90 degrees, 7-conductor			1		[4/]	See figure 7.
4	Plug assembly, 90 degrees, 9-conductor				1	[4/]	See figure 7.
5	Sleeve, plug	1	1	1	1	Ni-Cu, class A, QQ-N-281	See figure 3 and requirement 4.
6	Coupling	1	1	1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 4.
7	Cap, protective	1	1	1	1	Plastic, type MFI-20 or MAI-60, ASTM D5948 or polycarbonate, ASTM D3935 [6/]	Nonpressure-proof, see figure 5.
8	Sealing plug	1	1	1	1	Ni-Cu, class A, QQ-N-281	Pressure-proof, see figure 2 and figure 6.
9	Gasket, O-ring	1	1	1	1	SAE-AS28775-220	Lubricate with silicone compound, SAE-AS8660, see figure 6.
10	Disk, insulating	1	1	1	1	Non-laminated plastic, composition A, type II, grade GU, L-P-535	See figure 2.
11	Compound, polyurethane	As required				[7/]	See figure 1.
12	Cable	As required				[8/]	See figure 1.
13	Insert assembly, 7-conductor	1		1		M24231/29-004	See figure 1.
14	Insert assembly, 9-conductor		1		1	M24231/29-005	See figure 1.
15	Sealing plug, assembly [9/]	1	1	1	1	See items 8 and 9	See figure 6.
16	Compound, epoxy	As required				Cast epoxy resin compound [10/]	Base EE4183 or RE2039, hardener HD-3561; see figure 1 and figure 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ When plugs are purchased as a kit, the inserts are not molded in accordance with MIL-DTL-24231/29. The socket contacts and sleeving are individually provided.

5/ DELETED.

6/ May be Ni-Al-BRZ in accordance with ASTM B150/B150M at the option of the manufacturer.

7/ Molding compound to be procured in accordance with S9320-AM-PRO-020/MLDG.

8/ For cable type, see table for cross indexing of mating plugs, receptacles, adapters, and single and multiple cable connectors of MIL-DTL-24231.

9/ To be supplied when specified (see acquisition requirements in MIL-DTL-24231).

10/ Hysol base EE4183 or RE2039, hardener HD-3561, Hysol Division, Dexter Corporation, Olean, NY, CAGE 04347, or equal.

BACKWARDS COMPATIBILITY. Connectors with glass reinforced epoxy (GRE) plug sleeves procured to MIL-DTL-24231/3E, dated 26 April 2013, are not acceptable for shipboard use and should be disposed of. All future requisitions should be to the current revision of this specification sheet and MIL-DTL-24231.

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5935-2024-111)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:11Z
Check the source to verify that this is the current version before use.